Stage 8 Science — Practice Test 3 — Mark Scheme

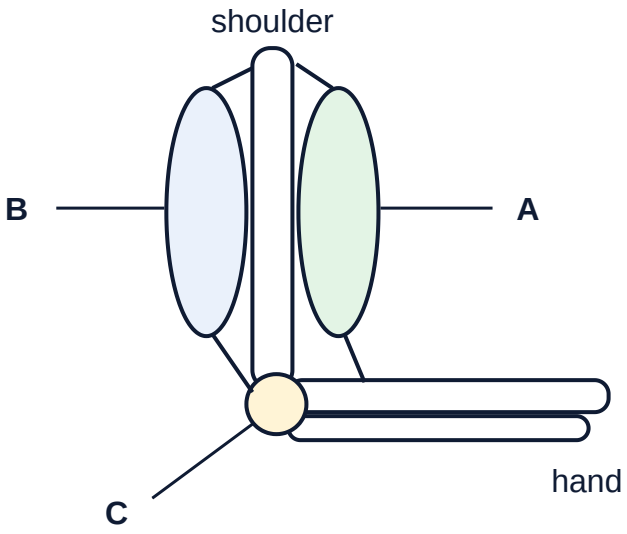
Paper 1: 45 minutes · calculator allowed · Paper 2: 45 minutes · Total 100 marks

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 45 minutes — 50 marks

1 Amir trains at a gym and wears a fitness tracker on his wrist. **[1]**

The diagram shows the bones and two muscles of his arm.



(a) Amir bends his arm at the elbow. What type of joint is the elbow?

- ☐ A) ball-and-socket joint
- ☐ B) fixed joint
- ☐ C) hinge joint
- ☐ D) a ligament

Answer: C) hinge joint
Notes / working: The elbow lets the forearm move backwards and forwards in one plane only, like a door on its hinges, so it is a **hinge joint**. A ligament is not a joint (it holds bones together at a joint).

(b) Amir can swing his arm round in a full circle at the shoulder. Name the type of joint at the shoulder. **[1]**

Answer: ball-and-socket or ball and socket or ball-and-socket joint or ball and socket joint
Notes / working: The shoulder is a **ball-and-socket joint**: the round end of the upper arm bone sits in a cup-shaped socket, so the arm can move in almost every direction, including in a circle.

(c) Amir lifts a dumbbell by bending his arm at the elbow. Which letter, **A** or **B**, shows the muscle that contracts? **[1]**

> **Answer:** A
> **Notes / working: A** (the biceps, on the front of the upper arm). It is joined to the forearm just below the elbow, so when it contracts (gets shorter) it pulls the forearm up and the arm bends. Muscle B (the triceps) relaxes at the same time.

(d) Explain why the muscles at a hinge joint must work as an **antagonistic pair**. **[2]**

> **Answer:** 1 mark: muscles can only **pull** (contract); they cannot push. 1 mark: so a second muscle is needed to pull the bone back the other way / when one muscle contracts the other relaxes. Accept: A bends the arm and B straightens it. Do not accept: 'the muscles push the bones'.
> **Notes / working:** A muscle can only do work by contracting and pulling on a bone — it cannot push. So one muscle (A) pulls the forearm up to bend the arm, and a second muscle (B) is needed to pull it back down to straighten it. When one contracts, the other relaxes: they work against each other, so they are 'antagonistic'.

(e) The fitness tracker shines light into Amir's skin to measure how much oxygen his blood is carrying. Which part of the blood transports oxygen? **[1]**

- ☐ A) plasma
- ☐ B) red blood cells
- ☐ C) white blood cells
- ☐ D) carbon dioxide

> **Answer:** B) red blood cells
> **Notes / working: Red blood cells** carry oxygen from the lungs to all the cells of the body. Plasma carries the blood cells, nutrients and carbon dioxide; white blood cells protect the body against pathogens.

2 Sara investigates how the speed of a treadmill affects her breathing rate. **[1]**

She runs for 3 minutes at each speed, then her fitness tracker counts her breaths in one minute. She repeats each speed three times on different days.

Speed in km/h	Test 1 (breaths per minute)	Test 2	Test 3	Mean
4	18	17	19	18
6	24	25	23	24
8	30	46	32	?
10	36	37	38	37

(a) Name the **independent** variable in Sara's investigation.

Answer: speed or speed of the treadmill or treadmill speed or the speed of the treadmill or running speed
Notes / working: The independent variable is the one Sara **changes** on purpose: the **speed of the treadmill**. The dependent variable (the one she measures) is her breathing rate.

(b) Give **one** variable Sara should keep the same to make her results valid. **[1]**

Answer: 1 mark for any one: the time she runs at each speed (3 minutes); the time the breaths are counted for (1 minute); the slope/incline of the treadmill; resting before each run / starting from the same resting breathing rate; same time of day / same temperature of the room. Do not accept: speed, breathing rate, 'the treadmill'.
Notes / working: A control variable is something that could also change the breathing rate, so it must stay the same: for example, how long she runs at each speed, how long she counts breaths for, or the slope of the treadmill.

(c) One result in the table is anomalous. Write down this result. **[1]**

Answer: 46 breaths per minute
Notes / working: 46 (test 2 at 8 km/h). The other two results at 8 km/h are 30 and 32, and 46 is much higher than them — it does not fit the pattern.

(d) Calculate the mean breathing rate at 8 km/h. Do **not** include the anomalous result. Show your working. **[2]**

Answer: 31 breaths per minute
Notes / working: Leave out 46. Mean = (30 + 32) ÷ 2 = 62 ÷ 2 = **31** breaths per minute.
Mark scheme: M1 for (30 + 32) ÷ 2 seen; A1 for 31. (If 46 is included, (30 + 46 + 32) ÷ 3 = 36 scores 0.)

(e) The results show that Sara breathes faster when she runs faster. **[2]**
Use ideas about **respiration** to explain why.

Answer: 1 mark: her muscles need more energy / respire faster (aerobic respiration). 1 mark: so they need more oxygen (taken in) and/or more carbon dioxide must be removed. Accept: glucose + oxygen → carbon dioxide + water happens faster. Do not accept: 'respiration makes energy' (energy is released, not made) — ignore this if the rest is correct.
Notes / working: Running faster means her leg muscles need more energy. The energy is released by aerobic respiration in the mitochondria (glucose + oxygen → carbon dioxide + water). More respiration needs more oxygen and produces more carbon dioxide, so she breathes faster to take in oxygen and remove carbon dioxide.

(f) Why did Sara repeat each speed three times? **[1]**

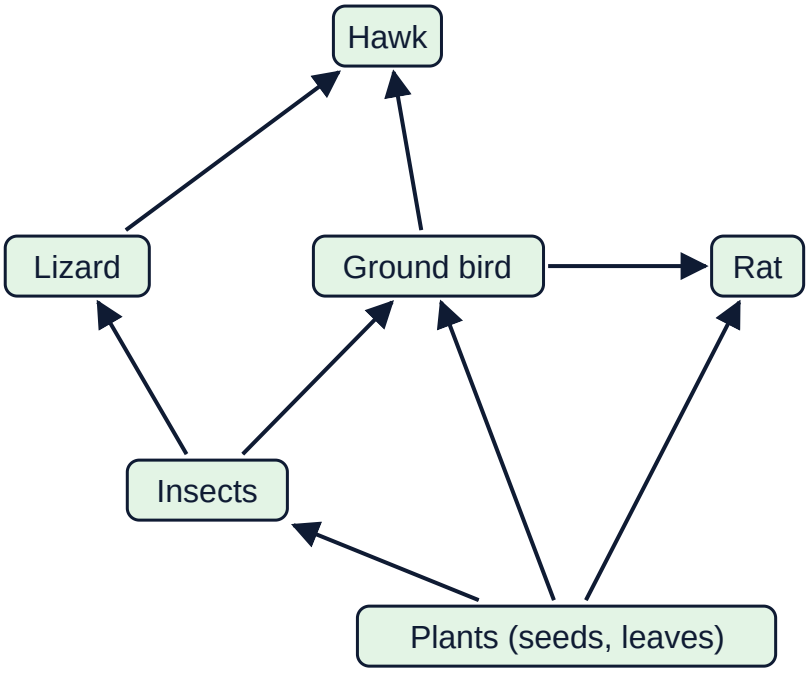
- ☐ A) to make the treadmill go faster
- ☐ B) to change the independent variable
- ☐ C) to have more control variables
- ☐ D) to spot anomalous results and make the results more reliable

Answer: D) to spot anomalous results and make the results more reliable
Notes / working: Repeating lets you see if one result is very different from the others (anomalous) and calculate a mean, so the results are more **reliable**.

3 A small island has a forest, a sandy beach and rock pools. **[1]**

Rats arrived on the island by accident on a boat. There were no rats on the island before. Rats eat seeds and also eat the eggs of ground birds.

The diagram shows part of the island's food web after the rats arrived.



(a) The island is an ecosystem. Name **one** habitat on the island.

Answer: forest or beach or sandy beach or rock pools or rock pool or the forest or the beach
Notes / working: A habitat is the place where an organism lives. The island ecosystem has several habitats: the **forest**, the **sandy beach** and the **rock pools** (any one).

(b) Rats are an **invasive species** on this island. What does this mean? **[1]**

- ☐ A) a species that has always lived on the island
- ☐ B) a species that only eats plants
- ☐ C) a new species brought to an area that spreads and harms the organisms already there
- ☐ D) a species that has died out

Answer: C) a new species brought to an area that spreads and harms the organisms already there
Notes / working: An invasive species is one that is **not native** — it has been brought to a new place (often by humans) — and it spreads quickly and harms the ecosystem.

(c) After the rats arrived, the number of ground birds went down. **[2]**
Use the food web to give **two** reasons.

Answer: 1 mark: rats eat the ground birds' eggs (so fewer young birds hatch). 1 mark: rats compete with the ground birds for food / seeds / plants. Do not accept: 'rats eat ground birds' unless eggs are mentioned; 'hawks eat the birds' (this was true before the rats came).
Notes / working: 1. Rats eat ground-bird eggs, so fewer chicks hatch and the population falls. 2. Rats and ground birds both eat plants (seeds), so they compete for the same food; there is less food for the birds.

(d) Scientists then found that the number of insects **increased**. Use the food web to explain why. **[1]**

Answer: 1 mark: there are fewer ground birds, so fewer insects are eaten. Accept: less predation on insects by ground birds. Do not accept: 'rats eat insects' (not in the food web).
Notes / working: Ground birds eat insects. With fewer ground birds, fewer insects are eaten, so the insect population grows.

---

4 A factory makes fertilisers for farms. **[1]**
One fertiliser, ammonium sulfate, is made by reacting ammonia with sulfuric acid. The reaction makes only one product.

(a) Complete the word equation.
ammonia + sulfuric acid → _____

Answer: ammonium sulfate or ammonium sulphate
Notes / working: The question says the reaction makes only one product, ammonium sulfate: ammonia + sulfuric acid → **ammonium sulfate**

(b) The solid that leaves the factory also contains some water and a little unreacted acid. Which statement describes this solid? **[1]**

- ☐ A) It is impure, because it is a mixture of more than one substance.
- ☐ B) It is pure, because it was made in a chemical reaction.
- ☐ C) It is pure, because it is a solid.
- ☐ D) It is an element.

Answer: A) It is impure, because it is a mixture of more than one substance.
Notes / working: A pure substance contains only one chemical. This solid contains ammonium sulfate, water and acid, so it is an **impure mixture**. Reactions do not always give a single pure product.

(c) A quality-control worker tests a 200 g sample of the dry fertiliser. It contains 190 g of ammonium sulfate. **[2]**
Calculate the percentage purity of the sample. Show your working.

> **Answer:** 95 %
> **Notes / working:** Percentage purity = (mass of the wanted substance ÷ mass of the sample) × 100
> = (190 ÷ 200) × 100 = **95%**
> Mark scheme: M1 for 190 ÷ 200 (× 100) seen; A1 for 95.

(d) The factory makes a second fertiliser by reacting potassium hydroxide (an alkali) with nitric acid. **[1]**
Complete the word equation.
potassium hydroxide + nitric acid → potassium nitrate + _____

> **Answer:** water
> **Notes / working:** An acid reacts with an alkali to make a salt and **water**:
> potassium hydroxide + nitric acid → potassium nitrate + water

---

5 On a space mission, tanks of helium gas are used to push fuel into the rocket engines. **[1]**
A helium atom has 2 protons, 2 neutrons and 2 electrons.

(a) What is the charge on a neutron?

- ☐ A) positive
- ☐ B) negative
- ☐ C) no charge
- ☐ D) it is positive in some atoms and negative in others

> **Answer:** C) no charge
> **Notes / working:** Protons are positive, electrons are negative and neutrons have **no charge** (they are neutral).

(b) What holds the electrons in the helium atom? **[1]**

- ☐ A) gravity between the electrons
- ☐ B) the electrostatic attraction between the positive nucleus and the negative electrons
- ☐ C) the magnetism of the neutrons
- ☐ D) friction between the particles

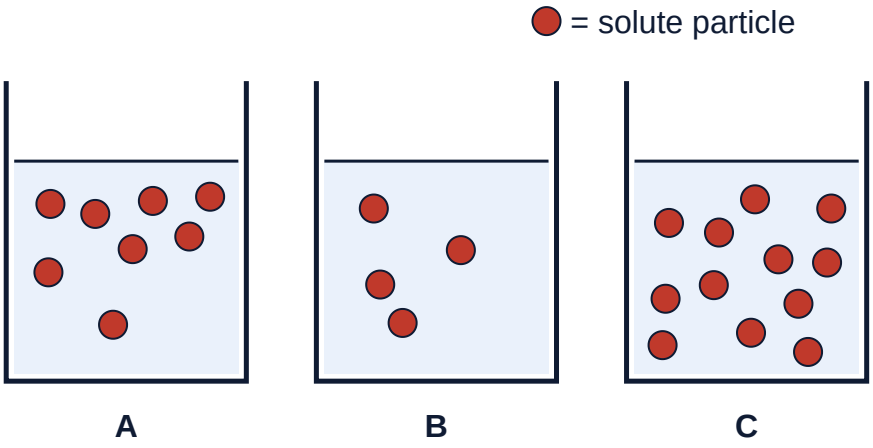
> **Answer:** B) the electrostatic attraction between the positive nucleus and the negative electrons
> **Notes / working:** Opposite charges attract. The **electrostatic attraction** between the positive nucleus (protons) and the negative electrons holds the atom together.

(c) Describe the Rutherford model of the atom. **[2]**

> **Answer:** 1 mark: a (tiny) nucleus at the centre that is positive / contains the protons (and neutrons) / contains almost all the mass. 1 mark: electrons move around (orbit) the nucleus / most of the atom is empty space. Do not accept: 'electrons are in the nucleus' or 'positive charge spread through the atom' (that is the plum pudding model).
> **Notes / working:** In the Rutherford model, almost all the mass and all the positive charge are in a **tiny nucleus** at the centre (protons and neutrons). The negative **electrons move around the nucleus**, and most of the atom is **empty space**.

---

6 The factory also sells a liquid fertiliser (a solution of a salt in water). **[1]**
The diagram shows the solute particles in three samples, A, B and C. Each beaker has the **same volume** of water.



(a) Which sample, **A**, **B** or **C**, is the most concentrated?

> **Answer:** C
> **Notes / working: C** has the most solute particles (12) in the same volume of water, so it is the most concentrated. (A has 8, B has 4.)

(b) A farmer adds more water to sample C. What happens to the concentration of the solution? **[1]**

- ☐ A) it decreases
- ☐ B) it increases
- ☐ C) it stays the same
- ☐ D) it becomes zero

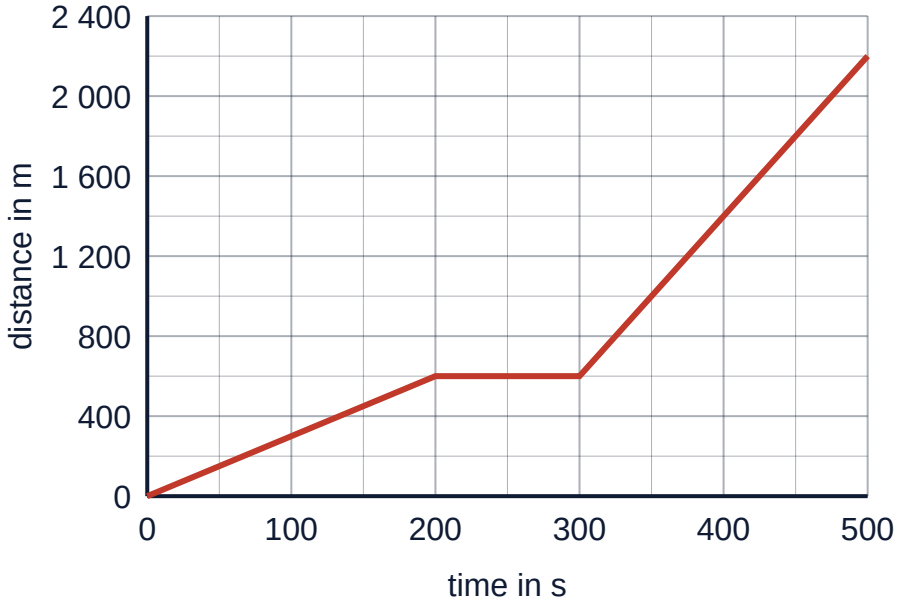
> **Answer:** A) it decreases
> **Notes / working:** It **decreases**: the solution becomes more dilute.

(c) Use ideas about particles to explain your answer to (b). **[1]**

Answer: 1 mark: the same number of solute particles is now in a larger volume of water / there are fewer solute particles in each cm^3 (in each unit volume). Do not accept: 'the particles get smaller' or 'particles are removed'.
Notes / working: Adding water does not remove any solute particles, but they are now spread through a bigger volume. There are fewer solute particles in each cm^3, so the concentration is lower.

---

7 Yusuf cycles up a hill, rests at the top, and then cycles down the other side. **[2]**
The distance–time graph shows his journey.



(a) Calculate Yusuf's speed going **up** the hill (the first 200 s). Show your working.

Answer: 3 m/s
Notes / working: From the graph he travels 600 m in 200 s.
speed = distance ÷ time = 600 ÷ 200 = **3 m/s**
Mark scheme: M1 for 600 ÷ 200; A1 for 3.

(b) Describe Yusuf's motion between 200 s and 300 s. **[1]**

Answer: stationary or stopped or not moving or at rest or he is stationary or he stopped or he is not moving or resting or he is resting or speed 0 or speed is zero or zero speed or speed = 0
Notes / working: The line is **horizontal**: the distance does not change, so he is **stationary** (resting at the top of the hill).

(c) During which part of the journey did Yusuf travel fastest? Explain how you can tell from the graph. **[1]**

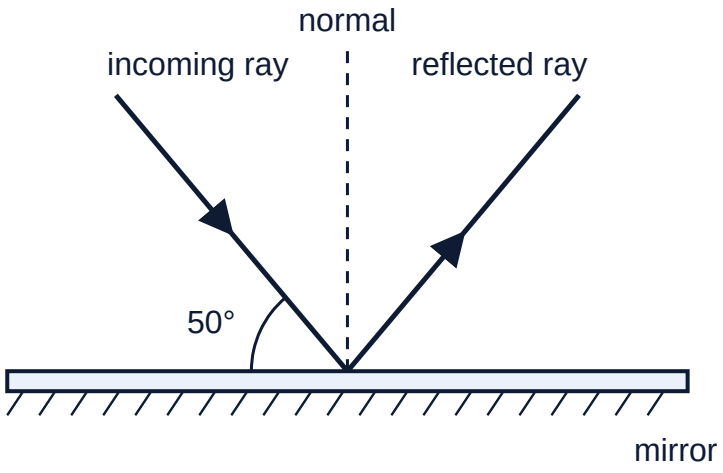
> **Answer:** 1 mark for both: 300 s to 500 s / going down the hill AND the line is steepest (has the greatest gradient/slope). Do not accept: 'the line is highest'.
> **Notes / working:** From 300 s to 500 s (going down the hill). The **steeper** the line on a distance–time graph, the faster the speed, and this part is the steepest (1600 m in 200 s = 8 m/s).

(d) Calculate Yusuf's **average** speed for the whole journey of 500 s. Show your working. **[2]**

> **Answer:** 4.4 m/s
> **Notes / working:** Total distance = 2200 m (read at 500 s); total time = 500 s (including the rest).
> average speed = 2200 ÷ 500 = **4.4 m/s**
> Mark scheme: M1 for 2200 ÷ 500; A1 for 4.4. Common mistake: leaving out the rest time, or averaging the three speeds (3, 0 and 8).

---

8 A theatre uses a flat mirror to send a spotlight beam onto the stage. **[1]**
The diagram shows a ray of light hitting the mirror. The angle between the incoming ray and the mirror is 50°.



(a) What is the **angle of incidence**?

> **Answer:** 40 °
> **Notes / working:** The angle of incidence is measured from the **normal**, not from the mirror. The normal is at 90° to the mirror, so angle of incidence = 90° − 50° = **40°**.

(b) What is the **angle of reflection**? **[1]**

> **Answer:** 40 °
> **Notes / working:** Law of reflection: angle of reflection = angle of incidence = **40°** (both measured from the normal).

(c) An actor wears a red costume. White light shines on it. **[2]**
Explain why the costume looks red.

> **Answer:** 1 mark: the costume reflects red light (into our eyes). 1 mark: it absorbs all the other colours (of white light). Do not accept: 'it makes red light' / 'it absorbs red'.
> **Notes / working:** White light contains all the colours. The red dye **reflects red light** into the audience's eyes and **absorbs all the other colours**, so we see red.

(d) The stage lights change so that only **blue** light shines on the red costume. What colour does the costume look now? **[1]**

- ☐ A) red
- ☐ B) blue
- ☐ C) magenta
- ☐ D) black

> **Answer:** D) black
> **Notes / working:** The red costume only reflects red light. Blue light has no red in it, so the costume absorbs the blue light and reflects nothing: it looks **black**.

(e) Two spotlights, one red and one green, overlap on a white screen. What colour is seen where they overlap? **[1]**

> **Answer:** yellow
> **Notes / working:** Adding coloured light: red + green = **yellow**.

---

9 At the fertiliser factory, salt crystals move along a conveyor belt. An electromagnet above the belt pulls out any small pieces of iron that have fallen into the salt. **[2]**

(a) Describe how to make a simple electromagnet.

> **Answer:** 1 mark: wrap/coil (insulated) wire around an iron core / iron nail. 1 mark: connect the coil to a cell / battery / power supply so that a current flows. Do not accept: 'rub the nail with a magnet' (that makes a permanent magnet).
> **Notes / working:** Wind a coil of insulated wire around an iron core (for example an iron nail), then connect the ends of the wire to a cell or power supply. When a current flows in the coil, the core becomes a magnet.

(b) Why does the factory use an electromagnet and not a permanent magnet? **[1]**

- ☐ A) it can be switched off, so the iron pieces drop off into a bin
- ☐ B) it also attracts the salt
- ☐ C) it never needs electricity
- ☐ D) it attracts copper and aluminium

Answer: A) it can be switched off, so the iron pieces drop off into a bin
Notes / working: An electromagnet is only magnetic while the current flows. The factory can switch it **off** to drop the collected iron in a bin. (Salt, copper and aluminium are not magnetic.)

---

10 A space agency is planning a mission to take pictures of a nearby galaxy. **[1]**

(a) Which list describes what a galaxy contains?

- ☐ A) only planets
- ☐ B) one star and the planets that orbit it
- ☐ C) only dust and gas
- ☐ D) billions of stars, planetary systems, and dust and gas

Answer: D) billions of stars, planetary systems, and dust and gas
Notes / working: A **galaxy** is a huge group of **stars** (many with **planetary systems**) together with clouds of **dust and gas**. One star with its planets is a planetary system (like the Solar System), not a galaxy.

(b) Before launch, engineers test the spacecraft's compass on Earth. The compass needle lines up with the Earth's magnetic field. **[1]**
Why does the Earth have a magnetic field?

- ☐ A) the Moon pulls on it
- ☐ B) the Earth's core acts as a magnet
- ☐ C) sunlight makes the air magnetic
- ☐ D) the oceans are magnetic

Answer: B) the Earth's core acts as a magnet
Notes / working: The Earth's magnetic field exists because the Earth's **core acts as a magnet** (the core contains iron).

11 Students in a climate data project collect information about their town. **[1]**

(a) Which statement describes the **climate** of the town, not the weather?

- ☐ A) It rained heavily this morning.
- ☐ B) Yesterday's highest temperature was 31 °C.
- ☐ C) Summers are usually hot and dry, based on 30 years of records.
- ☐ D) There is a strong wind today.

Answer: C) Summers are usually hot and dry, based on 30 years of records.
Notes / working: Weather is the conditions on a particular day or over a short time. **Climate** is the usual (average) pattern of weather over a long time, such as 30 years.

(b) The school's weather station gets its electricity from a solar panel. **[1]**
Is solar power a renewable or a non-renewable resource? Give a reason.

Answer: 1 mark for: renewable AND a reason, e.g. the Sun's energy will not run out / it is replaced naturally / it can be used again and again. Do not accept 'renewable' with no reason.
Notes / working: Solar power is **renewable**: energy from the Sun will not run out (it is replaced continuously), unlike fossil fuels.

Paper 2 — 45 minutes — 50 marks

1 The gym gives its members a leaflet about healthy eating. **[1]**

(a) Which nutrient is needed for strong bones and teeth?

- ☐ A) iron
- ☐ B) calcium
- ☐ C) vitamin C
- ☐ D) fat

Answer: B) calcium
Notes / working: Calcium (a mineral) builds strong bones and teeth. Vitamin D helps the body take in calcium. Iron is needed to make red blood cells.

(b) A weightlifter wants to build more muscle. Which nutrient is needed for growth and repair of the body? **[1]**

Answer: protein or proteins
Notes / working: Protein is used for growth and repair, for example making new muscle tissue.

(c) Animals can store energy in their bodies. Which **two** nutrients are used as energy stores? **[1]**

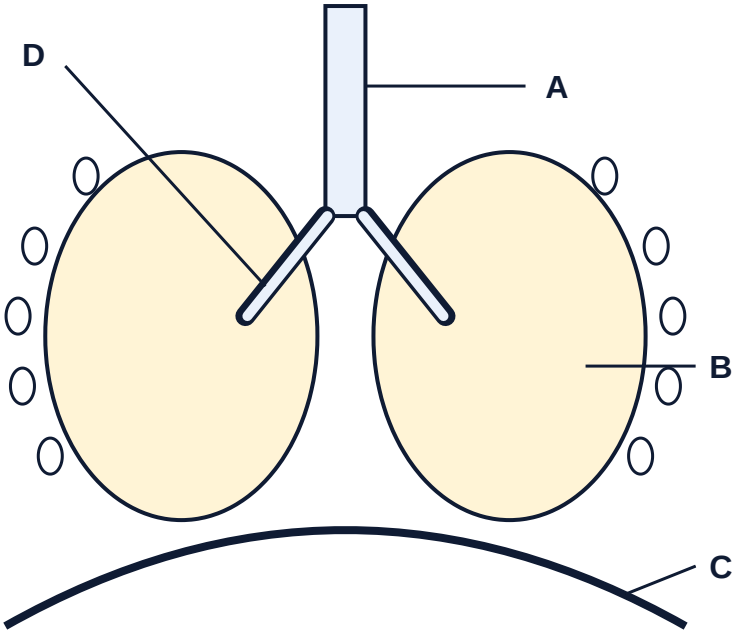
- ☐ A) vitamins and minerals
- ☐ B) protein and water
- ☐ C) calcium and iron
- ☐ D) carbohydrates and fats

Answer: D) carbohydrates and fats
Notes / working: Animals store energy as **carbohydrates** (e.g. glycogen in the liver and muscles) and **fats** (e.g. under the skin). They eat food to get energy and nutrients.

(d) The leaflet also warns gym members not to smoke. **[2]**
Describe **two** ways smoking can affect a person's health.

Answer: 1 mark each for any two: tar damages the lungs / causes lung cancer; damages the alveoli / causes breathing problems (e.g. bronchitis, emphysema); carbon monoxide reduces the oxygen the blood (red blood cells) can carry; causes heart disease / raises blood pressure; nicotine is addictive; person gets out of breath quickly / less fit. Do not accept 'it is bad for you' without detail.
Notes / working: Examples: tar in smoke can cause **lung cancer** and damage the alveoli, making breathing difficult; **carbon monoxide** in smoke means red blood cells carry less oxygen, so the person gets out of breath quickly; smoking also increases the risk of **heart disease**, and nicotine is **addictive**.

2 The diagram shows the human breathing system. A gym trainer uses it to explain what happens when we breathe. **[1]**



(a) Which letter shows the **diaphragm**?

Answer: C
Notes / working: C — the dome-shaped sheet of muscle under the lungs. (A is the trachea, B is a lung, D is a bronchus.)

(b) Which describes what happens when we breathe **in**? **[1]**

- ☐ A) the diaphragm relaxes and moves up; the volume of the chest decreases
- ☐ B) the diaphragm contracts and moves down; the volume of the chest increases
- ☐ C) the intercostal muscles relax and the ribs move down and in
- ☐ D) the diaphragm contracts and moves up; the volume of the chest decreases

Answer: B) the diaphragm contracts and moves down; the volume of the chest increases
Notes / working: Breathing in: the **diaphragm contracts and flattens (moves down)** and the intercostal muscles contract to pull the ribs up and out. The chest volume **increases**, the pressure inside falls, and air flows in.

(c) Describe how oxygen moves from the air in the alveoli into the blood. Use the word **diffusion** in your answer. **[2]**

> **Answer:** 1 mark: oxygen diffuses from a high concentration (in the alveoli) to a low concentration (in the blood). 1 mark: through the thin walls of the alveoli and capillaries / into the red blood cells. Accept: walls are one cell thick. Do not accept: oxygen is 'pumped' or 'sucked' into the blood.
> **Notes / working:** There is more oxygen in the air in the alveoli than in the blood arriving at the lungs. Oxygen **diffuses** from the high concentration in the alveoli to the low concentration in the blood, through the very thin walls of the alveoli and the capillaries, and into the red blood cells. (Carbon dioxide diffuses the opposite way.)

(d) A gym member says: "Breathing and respiration are the same thing." **[2]**
Explain why he is wrong.

> **Answer:** 1 mark: breathing is moving air into and out of the lungs (ventilation). 1 mark: respiration is a chemical reaction in cells / in mitochondria that releases energy from glucose (glucose + oxygen → carbon dioxide + water). Do not accept: 'respiration is breathing in and out'.
> **Notes / working: Breathing** is the movement of air in and out of the lungs, caused by the diaphragm and intercostal muscles. **Respiration** is a chemical reaction that happens in the mitochondria of every cell: glucose + oxygen → carbon dioxide + water, releasing energy.

---

3 On another island, farmers sprayed a pesticide on their fields. Rain washed some pesticide into the sea. **[1]**
Scientists measured the concentration of the pesticide in organisms in a sea food chain.

Organism	Pesticide concentration in units
algae	0.04
small shrimps	0.4
small fish	2
seabirds	20

(a) Which organism has the highest concentration of pesticide?

> **Answer:** seabirds or seabird
> **Notes / working:** The **seabirds** (20 units), at the top of the food chain.

(b) How many times greater is the concentration in the seabirds than in the small fish? **[1]**

> **Answer:** 10 times
> **Notes / working:** 20 ÷ 2 = **10** times greater.

(c) Explain why the concentration of pesticide increases along the food chain. **[2]**

Answer: 1 mark: the pesticide is not broken down / not excreted, so it stays (builds up) in the body of each organism. 1 mark: each animal eats many organisms from the level below, so it takes in all of their pesticide. Accept: the toxin is passed on and builds up at each level (bioaccumulation). Do not accept: 'the seabirds drink the sea water'.
Notes / working: The pesticide is not broken down or removed from the body, so it stays in each organism. Every shrimp eats lots of algae, every fish eats lots of shrimps, and every seabird eats lots of fish — so the pesticide from many organisms collects in one animal. This build-up is called **bioaccumulation**, and it can harm or kill top predators.

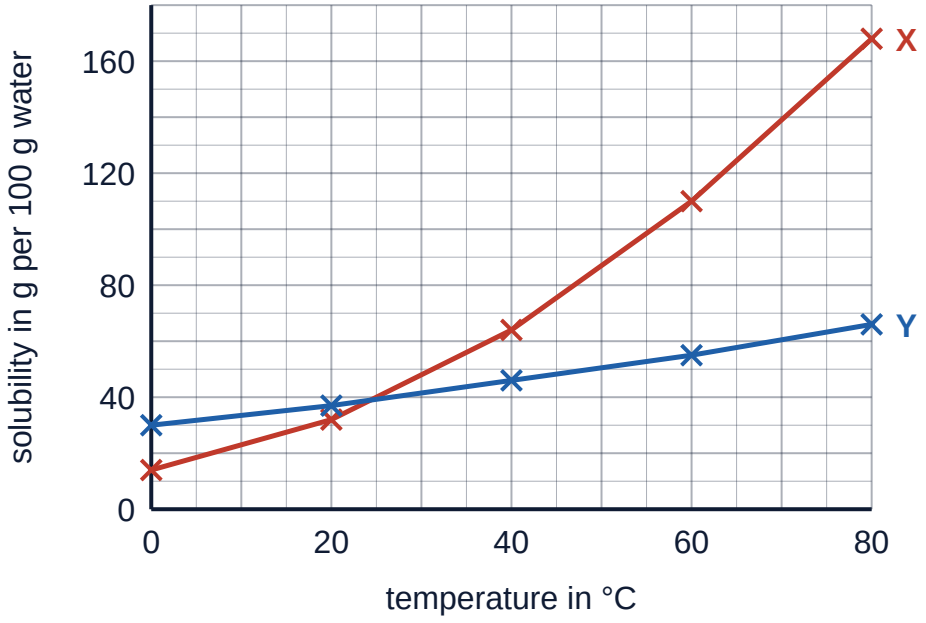
---

4 Scientists at the fertiliser factory investigate how the solubility of two salts, **X** and **Y**, changes with temperature. **[1]**

At each temperature they find the largest mass of salt that will dissolve in 100 g of water.

Temperature in °C	0	20	40	60	80
Salt X: solubility in g per 100 g water	14	32	64	110	168
Salt Y: solubility in g per 100 g water	30	37	46	55	66

The graph shows the results. The points are joined with straight lines.



(a) Name the **dependent** variable in this investigation.

Answer: solubility or the solubility or solubility of the salt or mass of salt that dissolves or mass that dissolves or mass of salt dissolved or mass dissolved
Notes / working: The dependent variable is what is **measured**: the **solubility** (the mass of salt that dissolves in 100 g of water). Temperature is the independent variable.

(b) Describe how the solubility of salt X changes as the temperature increases. **[1]**

Answer: 1 mark: the solubility increases (as temperature increases). Accept: it increases more and more quickly / the graph gets steeper. Do not accept: 'it changes'.
Notes / working: As the temperature rises, the solubility of X **increases**, and it increases faster at higher temperatures (the line gets steeper).

(c) Use the graph. At what temperature do salts X and Y have the **same** solubility? **[1]**

Answer: 24.5 °C
Notes / working: The two lines cross between 20 °C and 30 °C, at about **24 °C** (accept 22 °C to 27 °C). Below this temperature Y is more soluble; above it X is more soluble.

(d) A worker makes a saturated solution of salt X in 100 g of water at 60 °C. She then cools it to 20 °C. **[2]**
Calculate the mass of crystals of X that form. Show your working.

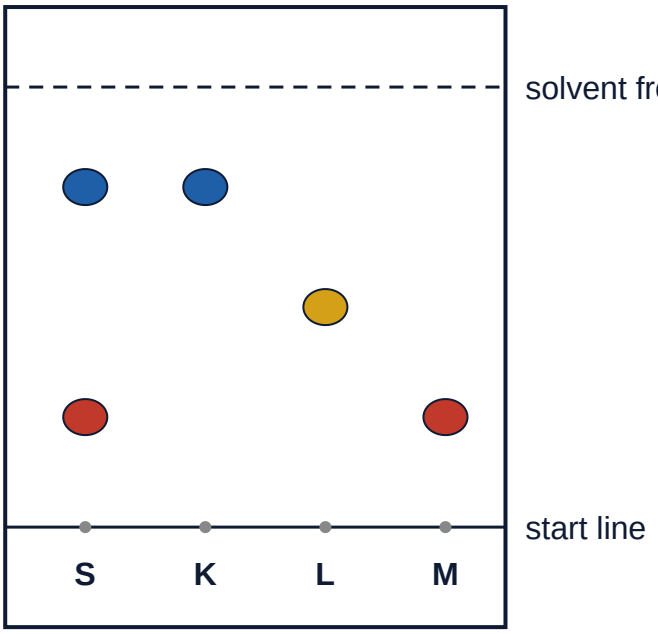
Answer: 78 g
Notes / working: At 60 °C, 110 g of X is dissolved. At 20 °C, only 32 g can stay dissolved.
Mass of crystals = 110 − 32 = **78 g**
Mark scheme: M1 for 110 and 32 read from the table/graph; A1 for 78.

(e) The scientists heat the water with a hot plate. Describe **one** risk and how to control it. **[1]**

Answer: 1 mark for a risk AND a matching control, e.g. burns from the hot plate/beaker → use tongs / heat-proof gloves / let it cool; hot liquid or salt splashing into eyes → wear eye protection (goggles); beaker knocked over → stand up / keep the bench clear. Do not accept a risk with no control.
Notes / working: For example: the hot beaker could burn your hand, so lift it with tongs or heat-proof gloves; or hot solution could spit into your eyes, so wear eye protection.

5 The factory adds coloured dyes to its fertilisers so that farmers can tell them apart. **[1]**

A scientist checks the dye in a fertiliser sample, **S**, using paper chromatography. She compares it with three known dyes, **K**, **L** and **M**.



(a) Which dyes are in sample S?

- ☐ A) K only
- ☐ B) K and L
- ☐ C) K and M
- ☐ D) L and M

Answer: C) K and M
Notes / working: S has two spots. One is at the same height as K and the other is at the same height as M, so S contains **K and M**.

(b) Explain why the start line is drawn in pencil, not in ink. **[1]**

Answer: 1 mark: ink would dissolve in the solvent / would separate and move up the paper and mix with the spots; pencil (graphite) does not dissolve. Do not accept: 'pencil can be rubbed out'.
Notes / working: Ink is made of dyes that dissolve in the solvent, so the line itself would move up the paper and spoil the results. Pencil (graphite) is insoluble, so the line stays put.

(c) What does the chromatogram show about sample S? **[1]**

- ☐ A) S is a pure substance
- ☐ B) S is a mixture of two dyes
- ☐ C) S contains dye L
- ☐ D) S does not dissolve in the solvent

Answer: B) S is a mixture of two dyes
Notes / working: A pure substance gives only **one** spot. S gives two spots, so it is a **mixture** of two dyes (it is not pure).

---

6 Ammonium nitrate is a fertiliser. A student dissolves some ammonium nitrate in water. **[1]**
The temperature of the water is 20 °C at the start and 12 °C at the end.

(a) By how many degrees does the temperature fall?

Answer: 8 °C
Notes / working: 20 − 12 = **8 °C**.

(b) Is dissolving ammonium nitrate **exothermic** or **endothermic**? **[1]**

Answer: endothermic
Notes / working: The temperature **falls**, so energy is taken in from the surroundings: the process is **endothermic**. (Exothermic changes give out energy and the temperature rises.)

(c) The factory engineer looks at how four metals react with cold water. **[1]**

Metal	Reaction with cold water
calcium	reacts steadily, giving bubbles of gas
copper	no reaction
magnesium	a few bubbles form very slowly
sodium	reacts very quickly, fizzes and moves about on the surface

Which list shows the metals in order of reactivity, **most** reactive first?

- ☐ A) sodium, calcium, magnesium, copper
- ☐ B) calcium, sodium, copper, magnesium
- ☐ C) copper, magnesium, calcium, sodium
- ☐ D) sodium, magnesium, calcium, copper

Answer: A) sodium, calcium, magnesium, copper
Notes / working: The faster the reaction, the more reactive the metal: sodium (very fast) > calcium (steady) > magnesium (very slow) > copper (no reaction). Answer: **sodium, calcium, magnesium, copper**.

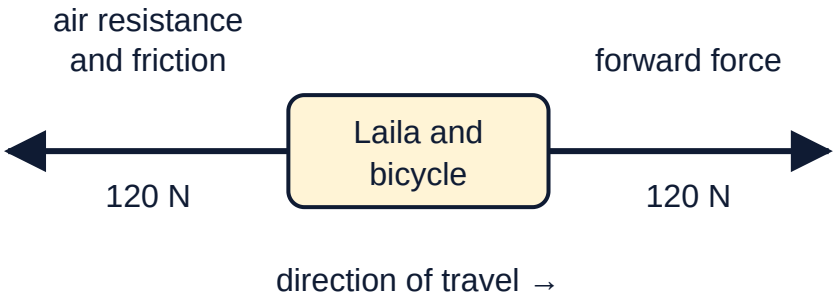
(d) Gold is used to coat electrical contacts in the factory's control room because it is **inert**. What does inert mean? **[1]**

- ☐ A) it reacts quickly with water
- ☐ B) it is magnetic
- ☐ C) it is generally unreactive
- ☐ D) it dissolves in water

Answer: C) it is generally unreactive
Notes / working: An **inert** substance is generally **unreactive**. Gold does not react with oxygen, water or dilute acids, so the contacts do not corrode.

---

7 Laila cycles to school. The diagram shows the horizontal forces on Laila and her bicycle on a flat road. **[1]**



(a) Laila is already moving. Describe her motion while these forces act.

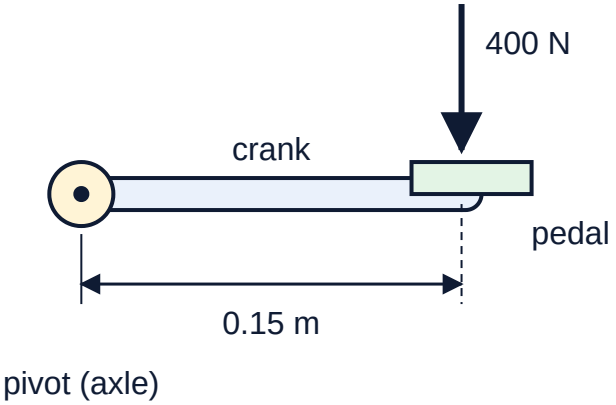
- ☐ A) she speeds up
- ☐ B) she slows down
- ☐ C) she stops at once
- ☐ D) she moves at a steady speed

Answer: D) she moves at a steady speed
Notes / working: The forces are **balanced** (120 N forwards, 120 N backwards), so the resultant force is zero. A moving object with balanced forces keeps moving at a **steady speed** in a straight line.

(b) At the bottom of a hill Laila pedals harder. The forward force becomes 150 N. The backward force stays at 120 N. **[1]**
Calculate the resultant force.

Answer: 30 N
Notes / working: Resultant force = 150 − 120 = **30 N** forwards. The forces are now unbalanced, so she speeds up.

[2]



(c) The diagram shows Laila's pedal crank. She pushes down on the pedal with a force of 400 N, 0.15 m from the pivot.
Calculate the turning moment. Show your working.

Answer: 60 N m
Notes / working: moment = force × perpendicular distance from the pivot = 400 × 0.15 = **60 N m**
Mark scheme: M1 for 400 × 0.15; A1 for 60 (N m). Common mistake: using 15 instead of 0.15 m gives 6000.

(d) Laila and her bicycle have a total weight of 800 N. The two tyres touch the road over a total area of 16 cm². **[2]**
Calculate the pressure on the road. Show your working.

Answer: 50 N/cm^2
Notes / working: pressure = force ÷ area = 800 ÷ 16 = **50 N/cm²**
Mark scheme: M1 for 800 ÷ 16; A1 for 50.

(e) Laila pumps more air into her tyre. Use ideas about particles to explain why the pressure of the air inside the tyre increases. **[2]**

Answer: 1 mark: there are more (air) particles in the same volume. 1 mark: so particles hit (collide with) the inside wall of the tyre more often / more collisions per second (so a bigger force on each area). Do not accept: 'the particles get bigger' or 'the particles expand'.
Notes / working: Pressure in a gas is caused by particles hitting the walls of the container. Pumping adds **more particles** into the same volume, so there are **more collisions with the tyre wall every second**, which means a bigger force on each cm² — a higher pressure.

8 During a show at a theatre, an actor sprays perfume on the stage. A lighting designer tests lights with a glass prism. **[1]**

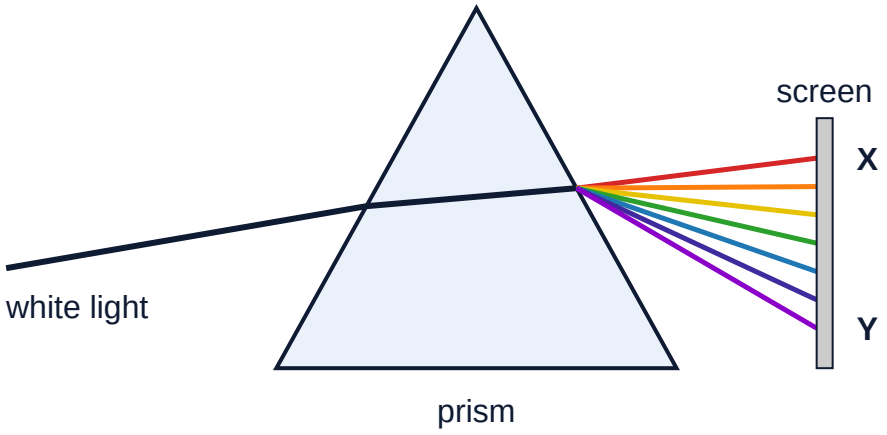
(a) After a few minutes, people in the front row can smell the perfume. Name the process by which the perfume particles spread through the air.

Answer: diffusion or diffuse or diffusing
Notes / working: Diffusion: the perfume particles move randomly and mix (intermingle) with the air particles, spreading from where there are many of them to where there are few.

(b) The designer shines a ray of light from air into a glass block. The ray bends as it enters the glass. **[1]**
Explain why light bends (refracts) at the boundary.

Answer: 1 mark: light changes speed / slows down as it goes from air into glass. Do not accept: 'the glass reflects the light' or 'the light changes colour'.
Notes / working: Light travels more **slowly** in glass than in air. The change of speed at the boundary makes the ray change direction (refraction).

[1]



(c) The diagram shows white light passing through the prism and making a spectrum on a screen. Name the colour seen at **Y**.

Answer: violet or purple
Notes / working: The prism bends (refracts) violet light the **most** and red light the least, so red is at X and **violet** is at Y. This splitting of white light into colours is called **dispersion** and shows that white light is a mixture of many colours.

(d) A yellow hat reflects red and green light. On stage, only **blue** light shines on the hat. What colour does the hat look? **[1]**

- ☐ A) yellow
- ☐ B) blue
- ☐ C) black
- ☐ D) green

Answer: C) black
Notes / working: The hat reflects only red and green light, and blue light contains neither. The hat absorbs the blue light and reflects nothing, so it looks **black**.

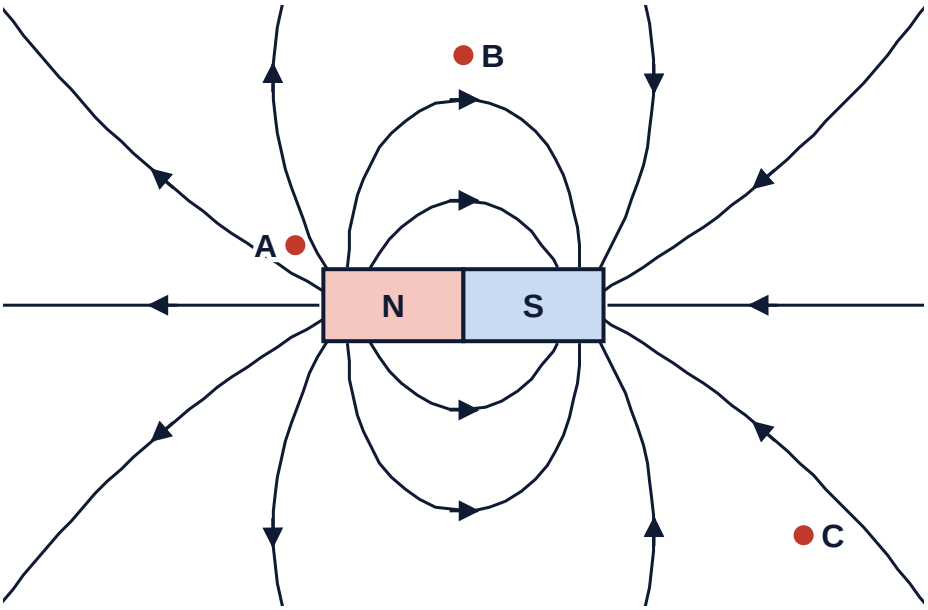
---

9 A spacecraft is travelling to study an asteroid. **[1]**

(a) Which statement about asteroids is correct?

- ☐ A) asteroids are rocks that are smaller than planets
- ☐ B) asteroids are small stars
- ☐ C) asteroids are larger than planets
- ☐ D) asteroids are made only of gas

Answer: A) asteroids are rocks that are smaller than planets
Notes / working: Asteroids are **rocks, smaller than planets**. They are rocks left over from when the planetary system (our Solar System) formed, which never joined together to make a planet.



[1]

(b) The spacecraft carries a bar magnet. The diagram shows the magnetic field around it.
At which point, **A**, **B** or **C**, is the magnetic field strongest?

Answer: A
Notes / working: **A**. The field is strongest near the **poles**, where the field lines are **closest together**. A is right next to the north pole; B and C are further away where the lines are spread out.

(c) Engineers test an electromagnet for the spacecraft. They change the number of turns on the coil and count how many paper clips it can hold. **[1]**

Number of turns	10	20	30	40
Number of paper clips	4	8	12	16

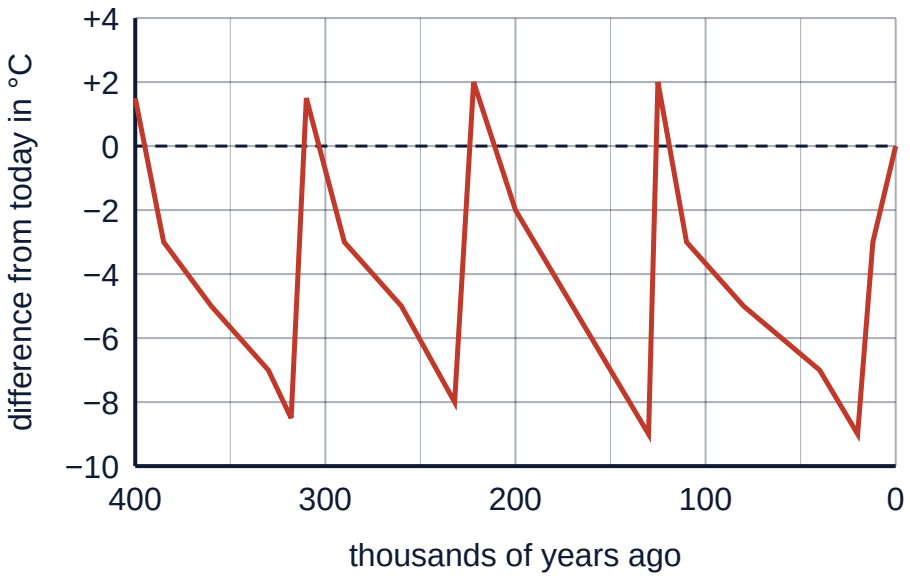
Predict how many paper clips the electromagnet will hold with 50 turns.

> **Answer:** 20 paper clips
> **Notes / working:** Every extra 10 turns holds 4 more clips (the number of clips is proportional to the number of turns). 40 turns → 16 clips, so 50 turns → 16 + 4 = **20** clips.

(d) Give **one** other way the engineers could make the electromagnet stronger, without changing the number of turns. **[1]**

> **Answer:** 1 mark for: increase the current / use more cells / a higher voltage; or use an iron core (if it has none) / a thicker iron core. Do not accept: 'use a bigger battery' unless linked to more current; 'use a copper core'.
> **Notes / working:** Increase the **current** in the coil (for example use more cells or a higher voltage), or make sure the coil is wrapped around an **iron core**.

10 In a climate data project, students study a graph made from ice-core data. It shows how much colder or warmer than today the Earth's average temperature was over the last 400 000 years. (The graph is simplified.) **[1]**



(a) Use the graph. About how many years pass from one warm period to the next? Give your answer in years.

Answer: 100000 years
Notes / working: The warm peaks are at about 400, 310, 220, 125 and 0 thousand years ago, so a warm period comes roughly every **100 000 years** (accept 70 000 to 130 000 years). The Earth's climate moves in a long cycle between warm periods and ice ages.

(b) What name is given to the long, very cold periods shown on the graph (about 8 °C colder than today)? **[1]**

Answer: ice age or ice ages or glacial or glacial period or glacial periods or glacials
Notes / working: Ice ages (glacial periods), when large ice sheets covered much more of the Earth.

(c) Scientists say the Earth's climate is warming now because the atmosphere is changing. Explain how burning fossil fuels can change the climate. **[2]**

Answer: 1 mark: burning fossil fuels releases carbon dioxide (a greenhouse gas) into the atmosphere. 1 mark: more greenhouse gas traps more heat (thermal energy) / less heat escapes into space, so the average temperature rises. Do not accept: 'it makes a hole in the ozone layer'.
Notes / working: Burning coal, oil and gas releases **carbon dioxide**, a greenhouse gas. More carbon dioxide in the atmosphere **traps more heat** near the Earth's surface, so the Earth's average temperature rises and the climate changes.

(d) Which of these energy resources is **non-renewable**? **[1]**

- ☐ A) wind
- ☐ B) tidal
- ☐ C) solar
- ☐ D) coal

> **Answer:** D) coal
> **Notes / working: Coal** is a fossil fuel: it took millions of years to form and will run out. Wind, tidal and solar power are renewable.

(e) A student finds a website about climate change that is paid for by an oil company. Explain why she should be careful when using information from it. **[1]**

> **Answer:** 1 mark: it may be biased — the company may want to show that burning oil is not harmful / benefits from selling oil. Accept: it may not give all the evidence; check other (scientific) sources. Do not accept: 'websites are always wrong'.
> **Notes / working:** The oil company makes money from selling oil, so the website may be **biased**: it might leave out evidence that burning fossil fuels changes the climate. She should compare it with other, independent scientific sources.